rogii-wellbore-geology-prediction

10th Place Solution: A Compass, Not a Map

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Team	Can
Author	Can
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10th place, solo: a compass, not a map

I spent ten weeks predicting where a drill bit sat inside the rock. The measurement that decided this competition for me was not a model. It was working out which of my own numbers was lying.

Result: 10th private, 21st public, 6,191 teams, solo entry.

Everything below is out of fold over **773 wells and 3,783,989 scored rows**. I rebuilt every number from the raw arrays before I believed it. The plotting script that made all the figures is at the end.

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1. Two facts that shaped everything

`TVT + Z` is a geological surface

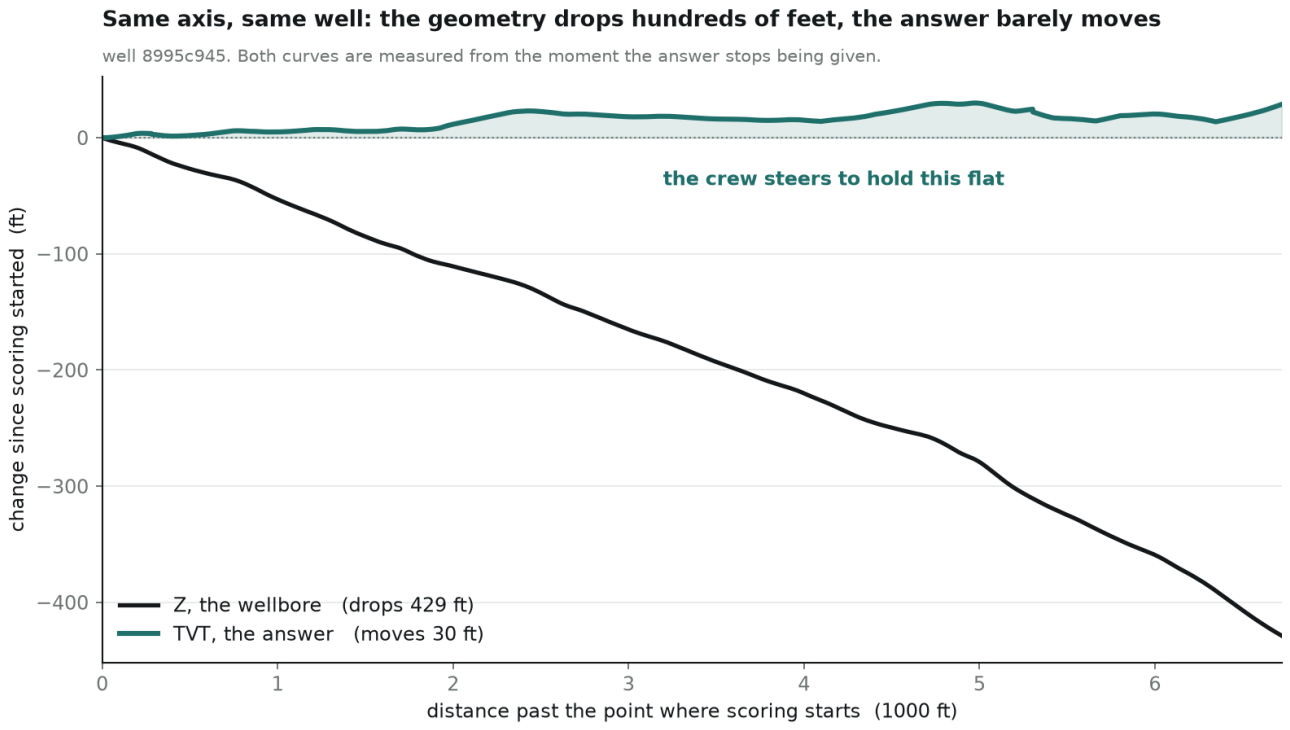
Take TVT, add Z, then subtract the formation top depth that the training set gives you. The result is **constant inside a well to 0.07 ft**, which is the rounding in the CSV. This holds for all 766 wells that have that column.

So the real object is:

$$A(MD) = TVT(MD) + Z(MD)$$

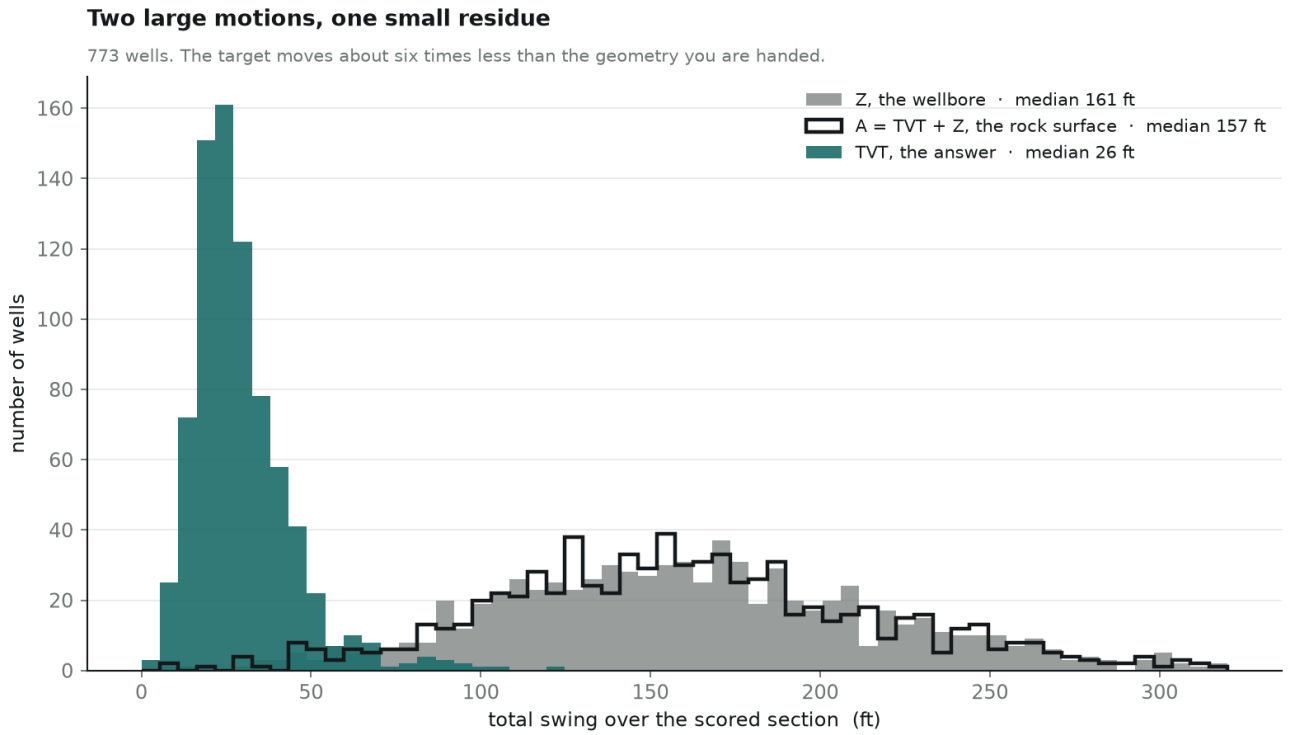
This is the height of a rock surface, sampled along the well path. Predicting TVT means extrapolating a smooth 3-D surface, then subtracting the Z you were already given.

The well steers, so the target barely moves



Over the scored part of a typical well:

quantity	median swing	p95
`Z`, the wellbore	**160.6 ft**	270.5
`A`, the rock surface	157.5 ft	266.1
`TVT`, **the answer**	**26.4 ft**	54.4



The crew is doing the work. They steer all the time to stay inside a layer that is moving. What you have to predict is the small difference between two large motions that almost cancel out.

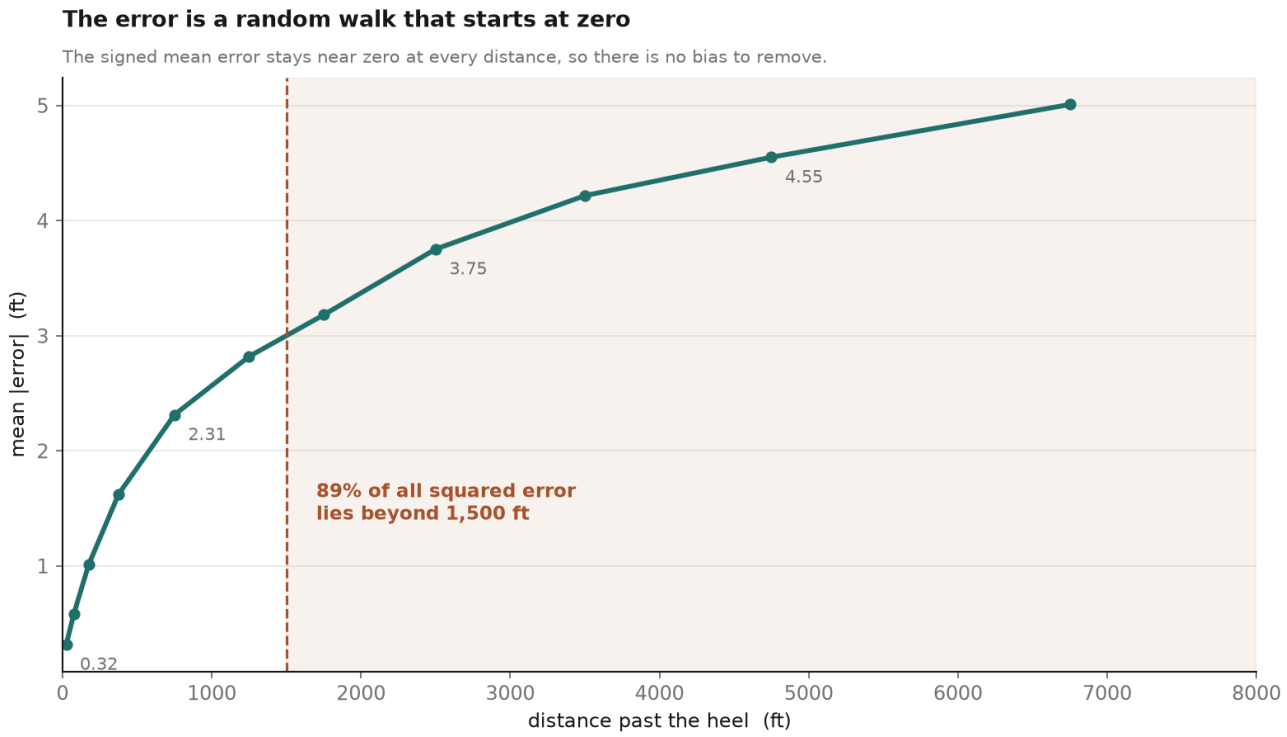
Why this matters for design. If you build a state space on `A`, you spend all your resolution on the 160 ft of geometry that you were given for free. If you build it on `TVT`, you spend it on the 26 ft you actually have to find. The smoothness prior still belongs on `dA/dMD`, the geological dip. I got this backwards twice and it cost me two working decoders.

Numbers for anyone building a lattice: `dA/dMD` p1 / median / p99 = -0.065 / +0.017 / +0.072. Standard deviation of `dTVT` per 20 ft node = 0.468 ft. Standard deviation of the dip change per node = 0.0096.

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2. Where the error lives

It grows like a random walk from the heel

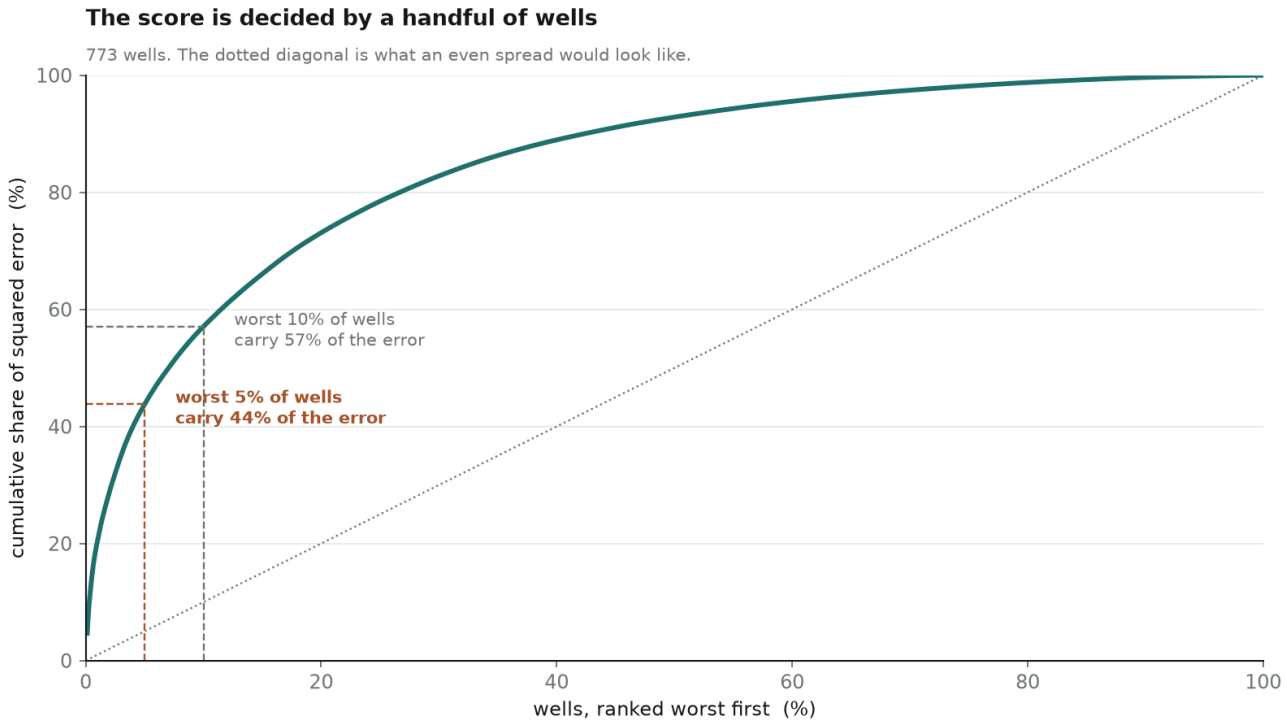


distance past the heel	first 50 ft	100 ft	500 ft	1,500 ft	3,000 ft	5,000 ft
mean |error| (ft)	0.32	0.60	2.31	3.18	4.20	**4.55**

The signed mean error is close to zero at every distance (-0.03 to -0.26). There is no bias to remove. There is only build up. After I remove a per well line, the residual autocorrelation is 0.9999 at lag 1, 0.8687 at lag 100, and only **0.1345 at lag 500**. So the walk has a correlation length of about 400 to 500 ft.

89% of all squared error sits beyond 1,500 ft from the heel.

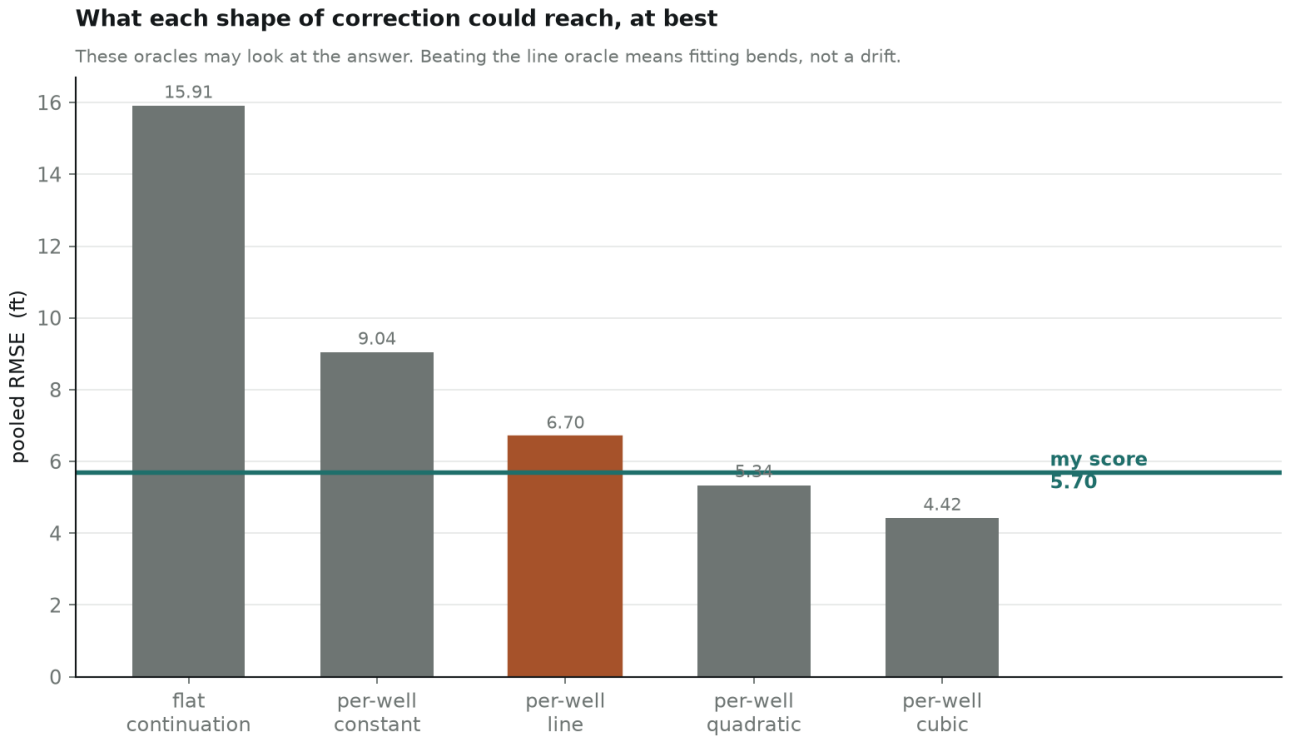
And it sits in very few wells



slice	share of total squared error
worst 10 wells (1.3% of rows)	22.7%
worst 39 wells (5%)	**43.9%**
worst 100 wells	62.4%

Per well RMSE at p5 / p25 / p50 / p75 / p95 = 1.22 / 2.14 / 3.31 / 5.62 / 11.08. If those 39 wells had only matched the median well, the pooled score would drop from 5.70 to about **4.39**. That is larger than every modelling gain of the whole competition put together, several times over.

What each shape of correction could reach



These oracles are allowed to look at the answer, so they tell you what a parameterisation is worth before you build it:

correction	pooled
flat continuation	15.91
per well constant	9.04
per well line	6.70
per well quadratic	5.34
per well cubic	4.42

On `A` with piecewise linear fits: one anchored line 7.59, one free line **5.96**, **two segments 2.43**, three segments 1.44.

My final 5.72 already beats the best possible single straight line. So anything below about 5.0 has to be fitting **bends**, not a drift. This killed the whole "just estimate the dip better" family before I wrote any of it.

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3. The solution, and what each stage was worth

The solution is a classical multi candidate engine with a learned selector, followed by two neural correctors.

Phase A, physics engine. All 773 training wells' formation top columns are subsampled into (X, Y, Z_surface) point banks with KD trees, plus a regional polynomial fit. A local plane and IDW interpolation gives a structural prior for `A` along the query well. Candidate curves come from a particle filter with 16 seeds and an HMM

forward backward pass over a stratigraphic position lattice, under both Gaussian and Cauchy likelihoods, with several datum mode variants.

Phase B, 2-D ranking U-Nets. Two families of 5 fold U-Nets read an image built from the lateral GR and the reference log, and produce a per row distribution over stratigraphic position. Fold logits are summed, then softmaxed.

Phase C, learned selector. A per well LightGBM scores every candidate curve using about 49 cross evidence features (agreement between engine variants, NN confidence, GR misfit statistics, lattice features). It turns the scores into weights with a tempered softmax and blends. A projection then pulls the blend back toward the structural prior.

Phase C2 and C3, neural correctors. A 23 channel per row tensor feeds a dilated TCN (20 checkpoints averaged), applied as `pred = blend + eta * clip(f, +-20)` with `eta = 0.10`. Then a second U-Net is mixed in.

stage	pooled CV	change	public
C1, engine + selector + projection	5.8120		5.729
+ TCN row corrector	5.7885	-0.024	
+ D2 network, scalar out of fold alpha	5.7544	-0.034	5.648
+ per row disagreement gated alpha	5.7244	-0.030	
+ heel anchored gain, s = 1.024	**5.7189**	-0.006	**5.620**

The last two rows are the ones I found in the final days. They are the only two gains I could show transferring: CV improved by 0.0355 and the public score improved by 0.028, on a different set of wells.

The gated alpha, because it is cheap and general. Instead of one blend weight for the second model, use a per row weight that depends on how much the two models disagree:

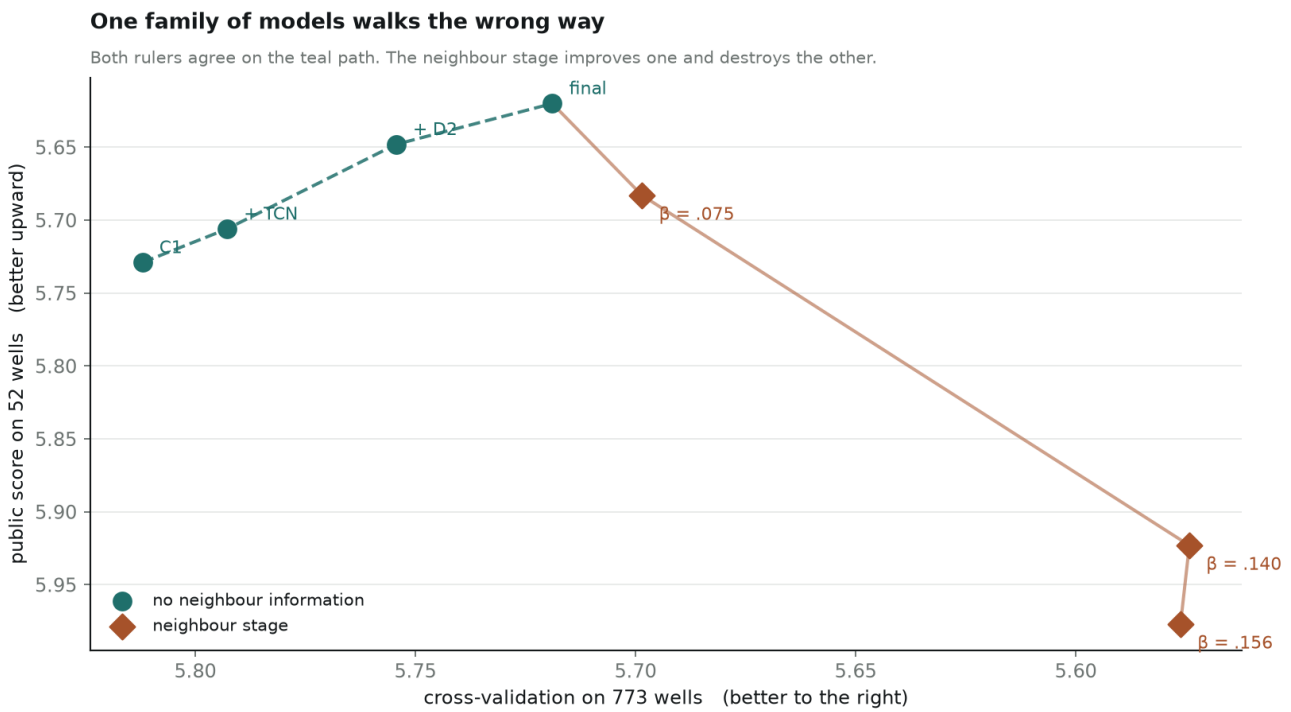
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alpha_i = clip(b0 + b1 * log1p|d2_i - chain_i|, 0, 0.6)
```

I fitted `(b0, b1)` by weighted least squares, out of fold. The values were stable across folds (b0 from 0.35 to 0.50, b1 from -0.08 to -0.14). Trust the second model where it agrees with the first, and back off where it does not.

A warning about reading these numbers. Per fold CV of the 5.7544 configuration was 5.50 / 5.99 / 5.65 / 5.51 / 6.09. That is a standard deviation of **0.28**, about five times the whole ladder. Every step above is smaller than the noise in any single fold. Only the paired comparison across all 773 wells is readable.

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4. The decision that actually decided it



Nearby wells share a geological surface. If a well 400 ft away has been interpreted all the way to its toe, its formation tops should constrain mine. I built that: a stage that anchors the chain to a plane fitted locally through nearby wells' interpreted tops.

It was the biggest single gain of the whole project. **CV went from 5.7544 to 5.5762, a jump of 0.18**, at a time when everything else was worth 0.03. Six independent rebuilds confirmed the arithmetic. A perturbation test confirmed it was not leaking the query well's own labels: destroying that well's own TVT and formation tops over the scored zone changed the candidate by exactly 0.000000 ft. Removing duplicate wells left the gain in place.

Then I submitted it. **Public went from 5.648 to 5.977.**

configuration	CV	public
no neighbour information	5.7544	5.648
neighbour stage, beta = .075	5.6985	5.683
neighbour stage, beta = .156	**5.5762**	**5.977**

Everything in me wanted to call this noise. It was not. On the correct ruler, where both configurations are scored on the same 52 public wells so the level cancels out, the damage is **2.95 standard deviations**, with an empirical p value of 0.002. All three high weight neighbour submissions land between 2.6 and 3.0 sigma. Their joint probability under noise is 0.0016. The three submissions without the neighbour stage all calibrate to within 0.2 sigma.

If I solve for the weight that the public score would have preferred, I get **beta = -0.032**. Negative.

The mechanism turned out to be measurable. The gain sits in wells that have a donor trace close by, and the hidden set is more isolated than the training set. Median distance to the nearest training well is **683 ft** for the

held out geometry against **470 ft** among training wells themselves. Pricing the stage under realistic donor density already removed a third of the gain (5.5762 to 5.6291).

> Cross validation was not wrong. It was answering a question about the training geometry, honestly and precisely. I was reading the answer as if it were about somewhere else.

I dropped the whole family. That decision, and not any model, is why this says 10th.

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5. The instrument: excellent locally, silent globally

The GR trace should tell you where you are. You match what the tool reads now against the vertical reference log, and the depth that matches is your position. I built that matcher and spent days sure it was working, because on ground where I could check it, it was.

Test 1, rigid localisation. Take a well's known section, where the true path is given, and ask the matcher to locate a rigid shift of that path. **81% of wells land within 1 ft.** The median error is 0.25 ft.

Test 2, matched perturbation. Perturb the true path with smooth, heel anchored deformations (correlation lengths 400, 800 and 1,600 ft, at 2, 5 and 10 ft of RMS) and ask the objective to rank the truth against them:

correlation length	amplitude	AUC, all rows	AUC, beyond 1,600 ft
400 ft	2 ft	0.884	0.857
400 ft	10 ft	0.947	0.903
800 ft	5 ft	0.911	0.873
1,600 ft	2 ft	0.871	0.847

By every local test I could build, the compass works. And it is worth almost nothing.

Test 3, retrieval. Around my chain prediction, I drew 1,025 smooth heel anchored candidates from an empirical, fold clean prior. The candidates are real residual shapes taken from wells in other folds and projected onto a 9 knot spline:

	pooled
the chain itself	6.01
best candidate in the set (oracle)	**2.60**
the candidate the objective picks	**7.09**
posterior weighted average	6.54

The answer is in the set. The objective cannot find it.

Telling two things apart is not the same as finding one thing among many. Even at 90% pairwise accuracy, the minimum over a thousand correlated candidates is decided by whichever one got lucky.

And the two skills do not predict each other. How sharply the compass located the truth in a well's known section predicts that well's final error with a rank correlation of **0.081**.

> I had a compass. The job needed a map.

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6. Fourteen directions, each closed with a number

direction	result
Extrapolate the drift from the known section	12.60 (flat = 15.91)
First fifth of the scored zone predicts the last	r = +0.131, slope r = +0.048
Borrow a neighbour's error	r = +0.025 within 2,000 ft
Per well offset from GR matching	r = -0.05, sign agreement at chance
Per well blend weight from 9 observables	5.769 against 5.744 for a fixed scalar
Second order decoder over (TVT, dip)	13.53 (flat = 13.34)
2-D CNN over the GR cost volume	14.5 (predicting zero = 15.80)
HMM forward backward, 9 parameterisations	11.77 (flat = 11.81)
Whole well residual matcher, U-Net + BiGRU	4.952 (predicting zero = 4.918)
Spline coefficients from the cost tube	r about 0 at every knot
Spline coefficients from trajectory geometry	r = -0.02 on average
Six fresh TCN seeds, ensembled	every mixture was worse
Detect the bad wells, then treat them	every treatment was worse
Neighbour stage limited to dense donor wells	gain scales with exposure, not density

Two of these deserve a sentence, because I expected both to work.

Seed ensembling

A top five competitor reported gaining 0.4 from equal weight ensembling. If seeds are exchangeable, then `MSE_K = C + (A - C) / K`, and a ten seed average needs a pairwise prediction RMS of about **2.34 ft** to give you that. It needs about **0.71 ft** to give you even 0.02. My six independently trained variants differ by **0.186 ft**. They were four times too similar for the smallest useful gain. His single model had a large removable random component. Mine did not. Measure your own diversity before you copy anyone's ensembling result.

Triage

Since 5% of wells carry 44% of the error, finding them before scoring should beat any model. The best leak free detector I built combines engine disagreement with the entropy of the matching cost. It reaches a rank correlation of 0.41 and catches 4 to 6 of the worst 39, where random would catch 2.

Then every treatment I applied to the flagged wells made things worse. Pulling an uncertain well back toward its anchor costs more than it saves, because the drift you are refusing is mostly real (a flat prediction scores 15.91 against my 5.70).

A hard well is not a kind of well. It is an outcome that builds up, and it is not written anywhere in the well's observable properties at the moment you would need to read it.

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7. The shake up, and why I ignored the public board

The public board scores 52 wells. The private board scores 148 different wells. I measured the paired standard deviation of a difference between two configurations at **0.168 on 52 wells and 0.093 on 148**. So a public gap below about 0.10 carries no information. I wrote that rule down before every submission and never let the public score break a tie.

The final board was harsher than I expected. The team that led the public board for weeks did not finish in the top ten. I moved from 21st to 10th without changing my judgement in the last days, only by spending them removing what a 52 well ruler had rewarded.

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8. Acknowledgements

Working alone means the discussion forum was my only window into what the top of the board was doing. Four people there changed how I spent my time, and it is worth being specific about how.

@tuckerarrants wrote publicly that a single non tabular model can reach a CV below 5, using only the well and its own reference log, with no neighbour well data at all. That comment did two things for me. Early on it pushed me toward neural sequence models while I was still adding tabular features. Much later, when my neighbour based stage improved cross validation by 0.18 and destroyed the public score, his remark was one of the reasons I believed the public score instead of the CV. If the top of the board did not need neighbour information, then my stage was probably reading the training geometry rather than the geology.

@ravaghi published the particle filter ensemble that the engine's filter and beam leg is a port of. It was my starting point for candidate generation and it stayed in the pipeline to the end.

Shrey Gandhi described his segment beam search in enough detail to reproduce, including the ablations for the persistence prior, the reference log recalibration and the drilled trend centering. It was the clearest specification anyone posted, and it shaped how I built my own decoder even though mine did not end up working.

Georgy Mamarin measured something I would not have measured myself: the spread of the public score across byte identical resubmissions, which is about 0.03. I used that number as a hard threshold for the whole competition. Any public difference below about 0.10 was treated as unreadable and never allowed to break a tie. That single measurement is a large part of why I survived the shake up.

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9. Coda: the right decision and the better outcome are not the same thing

One more number, because leaving it out would make this a worse document than the work deserves.

The neighbour stage I dropped, the one that gained 0.18 on CV and lost 0.33 on public, the one I killed at 2.95 sigma and never looked at again, scored **6.179** on the private set. My final selection scored **6.269**. Shipping the thing I spent a week proving I should not ship would have finished **9th**.

I would make the same call again. The evidence at decision time said the stage did not transfer, and it said so at three sigma across three separate submissions. A process that overrides that to chase 0.18 will be wrong far more often than it is right. But I am not going to dress up the outcome. Under uncertainty the correct decision and the better result come apart, and the honest thing is to report both and keep the process.

> What I learned is not a model. It is that my cross validation and the hidden set were answering two different questions, and that most of the skill here was noticing which one I was being asked.